LIBBY — PATIENT/CAREGIVER TRACK

gbm-mtap-cdkn2a-idhwt-subtotal-c4bq

Plain-language summary for patient and caregiver

Generated 2026-07-10

What this page is

This is the plain-language companion to a longer clinician page on the same case. It walks through what Libby found about a newly diagnosed glioblastoma, what tests need to happen before some decisions can be made, and what treatment options were ranked highest for a conversation with the neuro-oncology team. It is meant to help you arrive at appointments with sharper questions, not to replace anything your team says.

A few ground rules before reading. Libby looks at *targetable features* — specific molecular fingerprints of the tumor — and ranks options that act on those fingerprints, plus the standard-of-care treatment that has the strongest survival evidence. Several pending test results gate several of the options below, so a lot of this page reads as "if X comes back, then Y is on the table." No statistic here is a promise. Two of the options below (Options 7 and 8, both experimental trials in the MTAP drug class) were added after the expert panel finished its review, so the panel did not weigh in on them — that is flagged where each one appears.

What we know about the cancer

The tumor is a glioblastoma, IDH-wildtype, WHO grade 4 — the most common and most aggressive primary brain cancer in adults. "IDH-wildtype" means the tumor does not have a mutation in a gene called IDH1, which matters because IDH-mutant brain tumors get a different drug (vorasidenib) that does not apply here. Surgery was a subtotal resection, meaning some tumor was removed but not all of it — some enhancing tumor is presumed still there on the post-op MRI. The tumor had bleeding inside it at the time of diagnosis, which is a safety detail that carries forward through the rest of the page.

Two molecular features stand out:

- **MTAP loss.** MTAP is a gene on the short arm of chromosome 9. When it is deleted, the tumor cell starts piling up a metabolic byproduct called MTA. MTA partially blocks an enzyme called PRMT5. A new class of drugs has been engineered to take advantage of this — they finish the job MTA started, and they only really hurt the cancer cells (which already have the metabolic stress) rather than normal cells. This is the single most interesting molecular finding for trial-matching purposes.
- **CDKN2A loss (also called p16 loss).** CDKN2A sits right next to MTAP on chromosome 9, and in this tumor both were deleted together. CDKN2A normally puts the brakes on cell division; without it, the cancer cell divides more freely. This opens a separate drug class called CDK4/6 inhibitors, though the GBM-specific evidence for those drugs is more modest than the trial talk suggests (more on that below).

Several other tests have not come back yet, and the rest of the plan depends on them:

- **MGMT promoter methylation.** This is the single biggest test result still pending. MGMT is a DNA-repair enzyme; if its promoter is methylated (silenced), the tumor cell cannot easily undo the damage that the chemotherapy drug temozolomide causes. Methylated tumors get a lot of benefit from temozolomide; unmethylated tumors get much less. The standard treatment recommendation does not flip on this result (everyone still gets temozolomide), but the candor of the conversation about how much benefit to expect does.
- **EGFR amplification and EGFRvIII.** EGFR is a growth-signal receptor on the cell surface. About 40–50% of these tumors have extra copies of EGFR, and a fraction have a specific mutant form called EGFRvIII. EGFRvIII opens the door to some experimental cell-based therapies (CAR-T), which are mostly relevant later in the disease rather than now.
- **Comprehensive NGS panel.** This is a broad sequencing test that looks for rare but actionable findings —

BRAF V600E, NTRK fusions, microsatellite instability (MSI), and tumor mutational burden (TMB). The chance any of these come back positive in an adult IDH-wildtype glioblastoma is low (each one is roughly 1–2% or less), but if any of them does come back positive, an FDA-approved targeted drug becomes available. Since the panel is being run anyway, the cost of checking is essentially zero.

- **RB1 functional staining.** A confirmation that the cell-division brake pathway is intact downstream of CDK4/6, which is what would make the CDK4/6 inhibitor conversation worth having later.

The short version: this is a serious diagnosis. The standard treatment has real but modest survival evidence. The tumor has two unusual molecular features (MTAP loss and CDKN2A loss) that point at experimental drugs not yet approved, and there is one open trial that maps onto those features at a single US site.

What you told us matters most

The preference file says: surface every reasonable option, lean slightly toward efficacy over tolerability (0.7 out of 1), no modality vetoes, trials welcome including phase 1, and carry forward the bleeding history as a safety reason to avoid bevacizumab and any drug that thins the blood or attacks blood vessels. ECOG 1 (which means up and around, light activities, no significant restrictions) was assumed — confirm that with the neuro-oncology team.

The "surface all possible options" framing is what makes this page longer than a typical one. If priorities are actually narrower — keeping treatment close to home, avoiding repeat brain surgery, protecting cognition and quality of life over months rather than years, or staying off experimental drugs entirely — say that out loud at the first oncology visit. Those priorities will reshape the ranking below.

The first step everyone agreed on

Before any of the targeted options below can be properly ranked, several tests need to come back. None involve another biopsy or any toxicity. They all run on the tumor tissue that has already been collected from surgery, plus a single blood draw for the family-history piece. They can be ordered in parallel, and they take between five days and four weeks.

The tests are:

1. **MGMT promoter methylation (about 1–2 weeks).** Decides how much benefit to expect from temozolomide. Does not change the recommendation but changes the honest framing of it.
2. **Central MTAP staining by a specific antibody (about a week).** The trial at Ivy Brain Tumor Center in Phoenix requires this — they will not enroll on a tumor sequencing result alone. There is a story behind this: a different trial in the same drug class once enrolled four out of six patients on a sequencing-only call, and on confirmatory staining the tumors actually had intact MTAP. Those patients did not benefit. The lesson is to confirm before enrolling.
3. **EGFR amplification and EGFRvIII (about 1–2 weeks).** Opens or closes the cell-therapy conversation later.
4. **Comprehensive NGS panel with RNA fusion testing (about 2–4 weeks).** Looks for the rare actionable findings (BRAF V600E, NTRK fusions, MSI, TMB) that each have their own FDA-approved drug. The RNA piece is important — DNA-only panels can miss NTRK fusions in brain tumors.
5. **RB1 staining (about a week).** Confirms intact downstream machinery for any future CDK4/6 inhibitor conversation.
6. **Family-history germline panel (blood draw, 2–3 weeks).** If certain results turn up on the tumor side (Lynch syndrome genes, hereditary CDKN2A), it changes screening recommendations for first-degree relatives.

Practical advice: order all of these the week after the post-op MRI, in parallel, not one after the other. The radiation-start clock is short. Serialized testing eats up the window.

The options

Each option below is something for the neuro-oncology team to consider. They are ranked in roughly the order Libby's analysis weighted them, but rank order is not a prescription — it is a starting point. Several options depend on test results that are still pending.

Option 1 — Stupp regimen plus TTFields (Optune)

This is the standard-of-care treatment for newly diagnosed glioblastoma in adults under 70 with good performance status. It is what the major US, European, and global guidelines recommend as the preferred first-line treatment, and it is the regimen with the strongest survival evidence in this disease.

What it involves: six weeks of radiation to the tumor bed (60 Gy in 30 daily fractions) plus a low daily dose of temozolomide as a pill during those six weeks, then a four-week break, then six more cycles of higher-dose temozolomide pills (five days on, 23 days off, each cycle). Starting at the first higher-dose cycle, a wearable device called Optune is added — it sits against the scalp under transducer arrays and delivers a low-intensity alternating electric field (TTFields, 200 kHz) that disrupts the cancer cells' ability to divide. The catch with Optune is that the device only works if it is worn 18 hours a day or more. Below that, the survival benefit drops sharply.

The numbers from the registrational trials: with radiation plus temozolomide alone, median survival was about 15 months (versus 12 months with radiation alone), and 5-year survival was about 10% (versus 2%). Adding TTFields on top in the second trial pushed median survival to about 21 months, and 5-year survival to about 13%. These are not numbers anyone wants to be reading about a loved one, but they are the best a randomized phase 3 trial has produced in this disease, and the result has been replicated.

The MGMT result matters a lot for the temozolomide piece specifically. In MGMT-methylated tumors, temozolomide adds about 6–7 months of median survival. In MGMT-unmethylated tumors, the temozolomide-attributable benefit is smaller and the confidence intervals overlap. The radiation and the TTFields components keep working either way. The recommendation does not flip on the result — it is to give the regimen regardless — but the conversation about expected benefit is more honest if MGMT is known.

Main side effects: about 7% of patients have serious drops in blood counts during the radiation phase, rising to about 14% during the higher-dose pill phase. A specific lung infection (PJP) is a real risk during the radiation phase because of the immune suppression — preventive antibiotics are non-negotiable during that window. The scalp under the Optune arrays gets dermatitis in about half of patients (managed with array rotation and topical steroids); a small minority (2%) get serious enough scalp irritation to need a treatment change.

This is where the conversation starts.

Option 2 — BGB-58067 on the Ivy Brain Tumor Center trial (Phoenix)

This option is contingent on the MTAP staining coming back confirmed positive at central review. It also requires a willingness to travel and a second brain operation (more on that below).

BGB-58067 is one of a new generation of PRMT5 inhibitors that were engineered specifically to take advantage of MTAP loss. The trial (NCT07485049) is a phase 0/2 study running at one US site: Ivy Brain Tumor Center at St. Joseph's Hospital in Phoenix, with Nader Sanai as the lead investigator. The trial has two arms — one for MGMT-unmethylated tumors (drug plus radiation) and one for MGMT-methylated tumors (drug plus radiation plus temozolomide). Either arm keeps the standard regimen as the spine and layers the experimental drug on top — it does not replace standard care with the experimental drug.

What is unusual about the trial: the "phase 0" piece. Before the drug is given at full dose to a larger group, the trial does a short pre-treatment course followed by a second brain operation that pulls out a small amount of tumor tissue. The lab tests on that tissue confirm whether the drug is actually reaching the tumor and turning off the target. This is a smarter way to test a brain-cancer drug than the older approach of just guessing whether it crosses the blood-brain barrier, but it means signing up for a repeat surgery. That is the load-bearing practical question for this option.

What is contested about the option: the published evidence on BGB-58067 specifically is thin. The drug class is mechanistically clean (it is a precise match for this tumor's biology), and the chemistry was designed to fix a problem that killed an earlier drug in the same class (an earlier PRMT5 inhibitor was discontinued because it did not get into the brain at high enough concentrations). But the published safety and efficacy on BGB-58067 itself is not out yet. The closest comparison drug — a different molecule from a different sponsor — had a tumor-shrinkage rate of about 11% in heavily pre-treated patients with MTAP-loss tumors of various types, with a 14% rate of serious side effects and very few people stopping treatment because of toxicity. That is encouraging on safety but not a guarantee that this molecule will be the same. And that drug program was discontinued by its sponsor in April 2026, which is a real signal worth understanding before committing.

There is no published evidence yet that BGB-58067 helps people live longer. The argument for considering it is that the biology is an unusually clean match for this tumor, the trial keeps the standard regimen intact rather than replacing it, and there is no other open trial in the US right now that specifically targets MTAP loss in newly diagnosed glioblastoma. The argument against is that the molecule is unproven in humans for this disease, and committing to it means traveling to Phoenix and accepting a second brain operation.

Practical step: ask the neuro-oncology team to make initial contact with the Ivy Brain Tumor Center Phase 0 Navigator (contact details on the clinician page). Questions worth raising: travel-assistance fund availability, current slot openings in both arms, and the timeline for the published phase 0 safety readout. That readout, when it comes, will sharpen this conversation considerably.

Option 3 — Lomustine plus temozolomide plus radiation (CeTeG regimen)

This option is only available if MGMT comes back methylated. If it comes back unmethylated, this option is off the table entirely.

CeTeG is a German trial regimen that added a second chemotherapy drug (lomustine, a pill that crosses into the brain) to standard temozolomide plus radiation. In MGMT-methylated patients, the trial reported a median survival of 48 months versus 31 months on standard Stupp — a 17-month gain in the middle of the curve. If real, that is the biggest published survival signal in newly diagnosed glioblastoma.

The "if real" part is the caveat. The trial was small (141 patients total), the comparison was open-label (everyone knew who was on which arm), the analysis that produced that headline was per-protocol rather than intention-to-treat, and the confidence interval around the survival number crossed the line of "no effect" on the upper bound. The major US guidelines list the regimen as category 2A — meaning broadly accepted with uniform expert consensus, but on lower-quality evidence than the category 1 standard Stupp.

The toxicity is real. About 41% of patients have serious drops in blood counts on the doublet, compared with 13% on standard Stupp alone. At 66, cumulative blood-count damage from the second drug tends to bite by the third or fourth cycle, and many patients need dose reductions. There were no treatment-related deaths in the published cohort.

One important practical conflict: the BGB-58067 trial above (Option 2) does not include lomustine in its backbone. Committing to CeTeG outside of a trial would close the door to the methylated-MGMT arm of that trial. So if both options are on the table because MGMT comes back methylated, the choice between them is more than just a side-by-side comparison — it determines what else is available.

Option 4 — GBM AGILE platform trial

This is a multi-site adaptive platform trial (NCT03970447) that uses a smart randomization scheme — patients are assigned to one of several experimental arms, and over time the trial automatically routes new enrollees toward the arms that are doing best. It is enrolled at more than 40 US sites, so the geographic burden is much lower than the Phoenix trial above. Open enrollment is fast.

The catch: none of the current active arms is specifically targeted at MTAP loss. The two arms most likely to apply to a newly diagnosed glioblastoma patient are AZD1390 (an ATM kinase inhibitor that pairs with radiation) and tinostamustine (a combined HDAC inhibitor and alkylator). Both are mechanistically reasonable but neither is a biomarker match for this tumor's specific molecular features.

The structural argument for this option is its trial design — randomized, multi-site, with an active comparator that is current standard of care. That is a stronger experimental setup than a single-site phase 0/2. If the Phoenix logistics do not work or if the BGB-58067 slot is taken, this is the strongest randomized trial option still available.

The contact information for the platform sponsor (GCAR) is on the clinician page. Screening is fast.

Option 5 — SurVaxM (a tumor vaccine) on the SURVIVE trial

This option carries caveats and forces a binary choice with Optune. It is on this page because the safety profile is the lowest of any add-on option in the dossier, and the early signal from a small single-arm trial was encouraging.

SurVaxM is a peptide vaccine that targets a protein called survivin, which is overexpressed in glioblastoma cells. In a single-arm phase 2 trial of 63 patients, layered on top of standard Stupp, median survival was about 26 months versus a historical control of about 16 months. Side effects were essentially injection-site reactions and low-grade flu-like symptoms — no serious side effects attributed to the vaccine. That is the lowest-toxicity add-on in the entire dossier.

The "caveats" are two. The single-arm trial is not the same as a randomized one — those numbers need confirmation in a larger phase 2b randomized trial that is the validation gate. As of the latest verification, that trial is "active not recruiting," so it may or may not be open at any given enrolling site; the lead site (Roswell Park) is the place to check, and contact information is on the clinician page.

The second caveat is the binary choice: the SURVIVE protocol excludes Optune use. The patient cannot run

TTFields and SurVaxM together. So picking this option means giving up the second HR-0.63 survival signal that TTFields delivers on top of temozolomide — a known benefit with replicated phase 3 evidence — in exchange for an unreplicated single-arm vaccine signal. That trade is the honest framing of the option, and most reasonable people would not make it without much stronger randomized data than currently exists.

Option 6 — Tumor-agnostic targeted drugs (if the NGS panel turns up a positive result)

This is not a single option but a small family of "if X comes back positive, then Y is an option" branches. The comprehensive NGS panel from the workup step looks for three rare but actionable findings:

- **BRAF V600E** (about 1–2% prevalence in adult IDH-wildtype glioblastoma) opens enrollment for a two-pill combination called dabrafenib plus trametinib. In a trial in high-grade glioma with this mutation, about a third of patients had measurable tumor shrinkage.
- **NTRK fusion** (less than 1% prevalence) opens larotrectinib, a pill with a roughly 75–80% response rate across NTRK-fusion solid tumors and confirmed activity in brain tumors.
- **MSI-H or TMB-H** (uncommon in newly diagnosed IDH-wildtype glioblastoma; more likely later, after temozolomide exposure) opens pembrolizumab, an IV immunotherapy with about a 29% response rate in tumors with high TMB.

The prior probability that any of these comes back positive is low. But the NGS panel is being run anyway as part of the workup step, so the cost of checking is essentially zero, and the upside on a positive hit is large enough to reshape the second-line conversation. These are mostly relevant at recurrence — at first progression, not at diagnosis — but worth knowing about now.

A note on immunotherapy specifically: pembrolizumab and similar drugs have repeatedly failed in *unselected* glioblastoma — the published phase 3 trials in this disease are uniformly negative when the patient population is not pre-selected by biomarker. So pembrolizumab is only on this page through the narrow biomarker-gated door (MSI-H or TMB-H confirmed on the FDA-recognized platform), not as a general option.

Option 7 — Navlimetostat, a second brain-penetrant PRMT5 inhibitor

This option was added after the expert panel finished its review, so the panel did not weigh in on it. It is the same drug class as Option 2 — a PRMT5 inhibitor built to exploit MTAP loss — from a different sponsor (Bristol-Myers Squibb). What moved it onto the page is new data on one specific, hard problem: does the drug actually get into a brain tumor? Earlier drugs in this class failed on exactly that hurdle.

Here is what changed. There is now a dedicated brain-tumor trial (NCT06883747) running at the same Phoenix center as Option 2, and it is recruiting. It is built around a short course of the drug followed by a second operation that samples the tumor, so the lab can measure two things: whether the drug reached the tumor, and whether it switched off its target inside brain-tumor tissue. The first readout said yes to both — measurable drug levels in the non-enhancing part of the tumor, above the threshold the team had set in advance, and a strong drop in the marker that tells you the target is being shut off. In a separate cancer type (a lung cancer with the same MTAP loss), the drug shrank tumors in about 29% of heavily pretreated patients, with side effects that were on the lighter end for this class.

The honest framing matters here, because it is easy to over-read. The brain data so far is about the drug reaching and engaging its target — it is not proof that the drug shrinks brain tumors. No glioblastoma response data on this molecule has been published, and the lung-cancer shrinkage rate does not carry over to brain tumors; different disease, different odds. So what this data does is answer the "can the drug even get there" question that sank earlier drugs in the class. It does not answer the "does it help people with glioblastoma live longer" question. That one is still open.

Two practical limits. The recruiting brain-tumor trial requires the cancer to have come back and requires a planned re-operation, so a newly diagnosed patient is not eligible for it right now. And a separate cohort of the sponsor's larger trial (NCT07492680) is built for newly diagnosed glioblastoma with standard temozolomide and radiation — the same backbone-preserving design as Option 2, and multi-site, which would lower the travel burden a lot — but it is planned to open around July 2026 and is not enrolling yet. So neither piece is actionable today; both are worth pinning in the chart. The MTAP staining from the workup step is still the gate, and it is worth confirming whether these trials accept the same antibody or want a re-stain at their own lab.

If the Phoenix slot in Option 2 does not work out, this program is the strongest MTAP-axis backup to watch — with the caveat that the newly diagnosed cohort is a calendar trigger, not an option you can take this month.

Option 8 — TNG456 plus abemaciclib, a combination hitting both molecular changes at once

This option was also added after the expert panel finished its review, so the panel did not weigh in on it. It is the one option on the whole page that goes after both of this tumor's molecular changes at the same time. TNG456 is a brain-penetrant PRMT5 inhibitor aimed at the MTAP loss; abemaciclib is a CDK4/6 inhibitor aimed at the CDKN2A loss. Because those two deletions sit side by side on chromosome 9 in this tumor, a combination that hits both is, on paper, the most complete mechanistic match in the dossier. The trial (NCT06810544) is recruiting and lists glioblastoma as eligible.

The reason it sits this far down the list is that the evidence is at the earliest possible stage. So far there is laboratory data only — the combination shrank tumors in mouse models more deeply than either drug alone, and the PRMT5 piece was highly selective for MTAP-deleted cells — plus a trial that has just started enrolling. There is no human response data, no human safety table for the combination in glioblastoma, and the first patient was dosed only in mid-2025. So this is a promising idea backed by mechanism and animal data, not a treatment with a track record.

Two things to know. It is recurrence-gated: enrollment requires the cancer to have progressed, so it is a conversation for later, not now. And a combination of two drugs that can each lower blood counts has an unknown combined toxicity in this setting — that is a real open question, not a footnote. The CDK4/6 half of the combination also depends on the RB1 staining from the workup coming back intact; if RB1 is lost, that half of the rationale falls away.

Worth pinning for the recurrence conversation as the most mechanistically complete trial on the MTAP-plus-CDKN2A axis, with the plain caveat that it is unproven in people.

Option 9 — Abemaciclib (a CDK4/6 inhibitor) at first progression

This option carries caveats. It is on this page because the patient's tumor has the exact biology (CDKN2A loss with intact RB1) that this drug class was designed to attack, and the recurrence-time conversation deserves to be on the radar early. It is not a first-line option at any rank.

Abemaciclib is a twice-daily pill, FDA-approved for breast cancer, and used off-label in glioblastoma. The case for it in this tumor is mechanistic: CDKN2A loss removes the natural brake on cell division, and abemaciclib reinstalls that brake pharmacologically. Of the three available CDK4/6 inhibitors, abemaciclib was specifically selected for its (partial) ability to get into the brain — an earlier related drug (palbociclib) does not get into brain tumors at meaningful concentrations and failed when tested.

What is contested: the published evidence in glioblastoma is modest. A trial in recurrent CDKN2A-deleted glioblastoma reported that less than 10% of patients had their disease controlled at six months, and a second trial showed a marginal benefit on progression-free survival with no survival benefit. The preclinical case for combining a CDK4/6 inhibitor with a PRMT5 inhibitor (the same pairing Option 8 puts into a single trial) is intriguing — there is laboratory data showing deeper responses with the combination than either drug alone — but no clinical trial has read out that combination in glioblastoma yet. Off-label, off-protocol use also faces an insurance fight: the major drug compendium does not currently list glioblastoma as a supported off-label indication for abemaciclib, so a medical-exception letter would be needed.

Worth flagging for the recurrence conversation, not worth committing to before the cancer comes back.

Option 10 — CAN-3110 (an engineered cold-sore virus) at first progression

This is a "pre-position" option — an experimental treatment that becomes potentially relevant at recurrence, not now. Including it here as a heads-up so the relevant test can be ordered cheaply, in advance.

CAN-3110 is a modified version of the herpes simplex virus (HSV-1, the cold-sore virus) that has been engineered to infect and lyse glioblastoma cells. It is injected directly into the tumor at the time of a second brain operation. In a phase 1 trial in recurrent glioblastoma, median survival was about 12 months overall, but in the subset of patients who already carried antibodies to HSV-1 (because they had been exposed to the cold-sore virus earlier in life), median survival was about 14 months — roughly double the historical benchmark.

The practical step now is cheap: order an HSV-1 IgG blood test (about $20). The result roughly doubles the prior probability of benefit later if this option becomes relevant.

The trial (NCT03152318) is currently active but not recruiting at the lead site (Dana-Farber). Worth checking again at recurrence.

Option 11 — FUS-mediated blood-brain barrier opening (at first progression)

Another recurrence-time pre-position. FUS (focused ultrasound) uses sound waves directed through the skull to temporarily open the blood-brain barrier, which lets chemotherapy drugs reach the tumor at much higher concentrations. A small phase 1 trial showed 4–6× higher drug concentrations in the tumor with the device than without. The phase 3 trial (SONOBIRD, NCT04528680) is testing this with carboplatin in recurrent glioblastoma.

The practical step is to be in contact with a SONOBIRD-participating US site (Northwestern, Columbia, or MD Anderson) before recurrence rather than after, because the decision to implant the device has to be made at the operating table during the re-resection — not on a separate visit afterward. Calling Adam Sonabend's team at Northwestern is a reasonable starting point.

Option 12 — Front-line CAR-T (EGFRvIII-directed cellular therapy)

This option was considered but not recommended at this point in the disease. It is on this page because

Libby surfaces what was weighed and rejected, not just what was kept.

There is a small early-phase trial (CARv3-TEAM-E at Mass General, plus parallel programs at UCSF and Penn) that uses a patient's own immune cells, engineered to target a glioblastoma-specific mutant of the EGFR receptor, infused directly into the brain ventricles. The published phase 1 data were striking — three of three patients had near-complete tumor shrinkage on imaging within days — but two of three relapsed within one to two months. The trial is hypothesis-generating, not survival-validated, and the published cohort is single-digit patients.

The concern about using it at front-line in this specific patient is the bleeding history. The therapy can produce significant brain inflammation (cytokine release syndrome and ICANS-like neurotoxicity); none of the published phase 1 cohorts specifically included patients with prior intracranial bleeding, so the re-bleed risk under that kind of inflammation has not been characterized in the published evidence. Combined with the lack of survival data, the upfront use of this approach in this patient is not the right move.

This may re-open at recurrence if the EGFR/EGFRvIII testing comes back positive *and* if a future trial cohort publishes a safety analysis that specifically addresses re-bleed risk under cellular-therapy inflammation. Not before then.

A note on what is NOT on this page

Two things are deliberately not listed:

- **Bevacizumab.** A blood-vessel-blocking drug, FDA-approved for recurrent glioblastoma. Generally not on the table here because of the bleeding history at presentation — it carries a known risk of intracranial bleeding that compounds with prior bleeds. Not absolutely closed forever, but not a routine option in this case.
- **DCVax-L (a dendritic cell vaccine).** The expanded-access program for this product is closed to new enrollment, so it is not actionable.

Questions to ask the neuro-oncology team

1. Have we ordered all of the workup tests in parallel — MGMT methylation, the MTAP staining at central review, EGFR and EGFRvIII, the full NGS panel with RNA fusion testing, and the RB1 staining? When are the results expected, and when does radiation need to start relative to those results?
2. Once MGMT comes back, can we have a candid conversation about how much temozolomide is likely to help in this specific case? Either way, the standard regimen is the recommendation — but the expected benefit conversation depends on that result.
3. For the Optune device: what does the 18-hour-a-day wear-time look like in practice for a 66-year-old who wants to keep daily life as normal as possible? Is the survival benefit worth the daily commitment, in our judgment?
4. If the MTAP staining confirms, how do we feel about the Phoenix trial? Specifically: would a second brain operation be feasible given what surgery already showed about the tumor? Is the travel realistic? What is the latest on the published BGB-58067 phase 0 data?
5. If MGMT comes back methylated, where do you land on adding lomustine (the CeTeG regimen)? The 17-month median survival gain is striking but the trial was small and the confidence interval crosses zero — how does that read in this case?
6. The two newer MTAP-class trials (Options 7 and 8) were added after the expert panel finished, so it did not review them — one where a related drug just showed it reaches a brain tumor, and one combining two drugs

against both molecular changes. Are either worth tracking now, and what would trigger a real conversation about them: a published readout, a nearer-to-home site opening, or the cancer coming back?

7. If the NGS panel comes back negative for all the rare actionable findings (which is the most likely outcome), and the MTAP staining does not confirm, what is left on the menu beyond standard Stupp plus Optune? At that point, what does the post-recurrence plan look like?
8. Is there a SONOBIRD-participating site we should be in contact with now, before recurrence, so that the focused ultrasound option is in play if and when first progression happens? And should we order the HSV-1 antibody test now to know whether CAN-3110 would be a stronger option later?
9. Are there genetic counseling or family-history considerations from the germline panel that we should be thinking about for our family?

Where things stand, honestly

This is a hard diagnosis. The standard treatment (radiation plus temozolomide plus Optune) is the regimen with the strongest randomized evidence in the disease, and it remains the recommendation here, but the median survival numbers from those trials — around 21 months — are sobering. The tumor has two unusual molecular features that point toward a new class of experimental drugs (PRMT5 inhibitors), and there is a single open US trial that fits this case well, at the cost of travel to Phoenix and a second brain operation. The same drug class has a couple of newer parallel tracks worth watching — one where a related drug has just shown it actually reaches a brain tumor (a hurdle that sank earlier drugs), and one that goes after both of this tumor's molecular changes at once — but those are early, mostly gated on the cancer coming back, and unproven in people so far. Several other "if the test comes back X" branches exist, mostly low-probability but worth checking because the test is being run anyway. The pending results from the workup step shape almost everything else on the page, so the most important thing right now is to get all of those tests ordered in parallel, this week. The rest of the conversation gets sharper once they come back.

Sources

The recommendations on this page draw on the following publications and clinical trials:

- Stupp regimen (EORTC 26981 / NCIC CE.3, registrational trial): PMID 15758010 (efficacy), PMID 15758009 (MGMT subgroup), PMID 19269895 (5-year update)
- Hegi MGMT biomarker analysis: PMID 15758010
- TTFields plus temozolomide (EF-14): PMID 29392280
- Perry CCTG CE.6 elderly hypofractionated radiation: PMID 28296618
- CeTeG / NOA-09 (lomustine plus temozolomide): PMID 31488360
- BGB-58067 at Ivy Brain Tumor Center: NCT07485049
- Anvumetostat MTAPESTRY-101 (PRMT5 class proxy): PMID 39282709
- GBM AGILE platform: NCT03970447
- Navlimetostat (BMS-986504) brain-tumor CNS-penetration trial: NCT06883747; newly diagnosed GBM cohort: NCT07492680
- TNG456 plus abemaciclib combination trial: NCT06810544
- SurVaxM phase 2a: PMID 36473145; SURVIVE phase 2b: NCT05163080
- Dabrafenib plus trametinib in BRAF V600E high-grade glioma (ROAR): PMID 34838156
- Larotrectinib (tumor-agnostic NTRK fusion): PMID 29466156
- Pembrolizumab in TMB-high (KEYNOTE-158): PMID 32919526

- Abemaciclib in recurrent CDKN2A-deleted glioblastoma: PMID 37722087
- Palbociclib in recurrent glioblastoma (cautionary precedent): PMID 30151703
- CAN-3110 oncolytic HSV-1 phase 1: PMID 37853118; trial: NCT03152318
- SonoCloud-9 / SONOBIRD phase 1 and 3: PMID 37087102; NCT04528680
- CARv3-TEAM-E (Choi 2024 NEJM phase 1): PMID 38477966; Penn dual-target (Bagley 2024 NatMed): PMID 38497615; Goff 2019 (treatment-related death precedent): PMID 30418325; trials: NCT05660369, NCT06186401, NCT07209241